

Three Charged Lepton Generations and the Koide Ratio from Holographic Boundary Encoding

A conditional derivation from a virial partition ansatz in the Informational Actualization Model

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Abstract

The Koide relation for charged leptons, $Q \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}$, has held at the 10^{-5} level for decades and lacks an accepted Standard Model derivation. This note develops a compact conditional argument: *if* (i) charged-fermion mass encodings on a holographic screen reduce, in the lowest-energy sector, to a single Fourier mode on a $U(1)$ charge orbit $S^1 \subset S^2$ and (ii) the IAM “virial partition” identifies the event-like and configuration-like contributions of that encoding with the kinetic and potential halves in the virial theorem for a $1/r$ interaction, then two statements follow with no further parameter fitting: (1) the number of admissible equally spaced phase states with positive mass is uniquely $n = 3$; and (2) for $n = 3$ the Koide ratio is an exact algebraic identity. The paper is written to be referee-facing: all nonstandard inputs are isolated as explicit assumptions, the logical dependence is made fully transparent, and the scope and failure modes are spelled out.

1 Executive summary (one page)

What is proven here (conditional theorems). Assuming a simple boundary encoding

$$\sqrt{m(\varphi)} = x + y \cos \varphi, \quad x > 0, \quad (1)$$

with n generations given by Z_n phases $\varphi_k = 2\pi k/n$, and assuming the virial constraint

$$\frac{y}{x} = \sqrt{2}, \quad (2)$$

we show: (i) mass positivity excludes $n = 2$ and all $n \geq 4$, leaving $n = 3$ as the unique nontrivial option; and (ii) for $n = 3$, the Koide ratio equals $2/3$ identically (no approximations). These results are purely algebraic once (1)–(2) are adopted.

What is *not* proven here. This paper does *not* derive the holographic dictionary for de Sitter space, nor does it derive (1) or the virial-to-encoding identification from quantum gravity. Instead, it isolates the minimal assumptions required for the conditional derivation. A referee can therefore assess the claim at the correct level: “If these inputs are granted, the output follows.”

Why write a conditional paper at all? Because the Koide identity is extremely rigid: any mechanism that *forces* a Z_3 phase structure on square-roots of masses automatically explains the relation. The value in a short note like this is to supply a physically motivated route to that Z_3 structure that is logically clean and falsifiable.

2 Background and scope

The Informational Actualization Model (IAM) is a gravitational-decoherence framework developed in Refs. [1–3, 5]. In its cosmological sector, IAM introduces a late-time matter-sector modification through an effective Hubble-friction term and a derived coupling $\beta_m = \Omega_m/2$ (no background modification). In its particle-physics sector, IAM treats stable particles as self-consistent “decoherence loops” whose rest energies can be related to holographic encoding costs (e.g. Ref. [5]).

This paper focuses only on a *kinematic* question: *Given a minimal boundary-encoding ansatz and a virial partition condition, what follows for the number of positive-mass phase states and for Koide’s ratio?* No claims are made about renormalization group running, QED/QCD corrections, or a UV-complete holographic dual.

3 Visualization and reproducibility assets (supplemental)

The repository accompanying this manuscript includes a small set of non-essential visualization assets to aid intuition and to make the mapping from the Z_3 phase construction to standard geometric interpretations easy to verify. These figures are *illustrative* only; no claim in the paper depends on them.

An accompanying animation (`koide_slider_animation.gif`) is provided as a repository supplement; it is not intended for inclusion in the compiled PDF.

4 Assumptions

We state the nonstandard inputs explicitly.

Assumption 1 (Charge-orbit reduction to a circle). *A nonzero bulk $U(1)$ charge corresponds, on the holographic screen, to a one-parameter orbit S^1 (a “charge fiber”) on which the relevant phase variable φ lives. All states with the same absolute charge $|Q| = 1$ are supported on the same orbit.*

This is the minimal piece of the usual AdS/CFT dictionary (bulk gauge $\leftrightarrow$ boundary global symmetry) that we import; its de Sitter status is discussed in Appendix A.

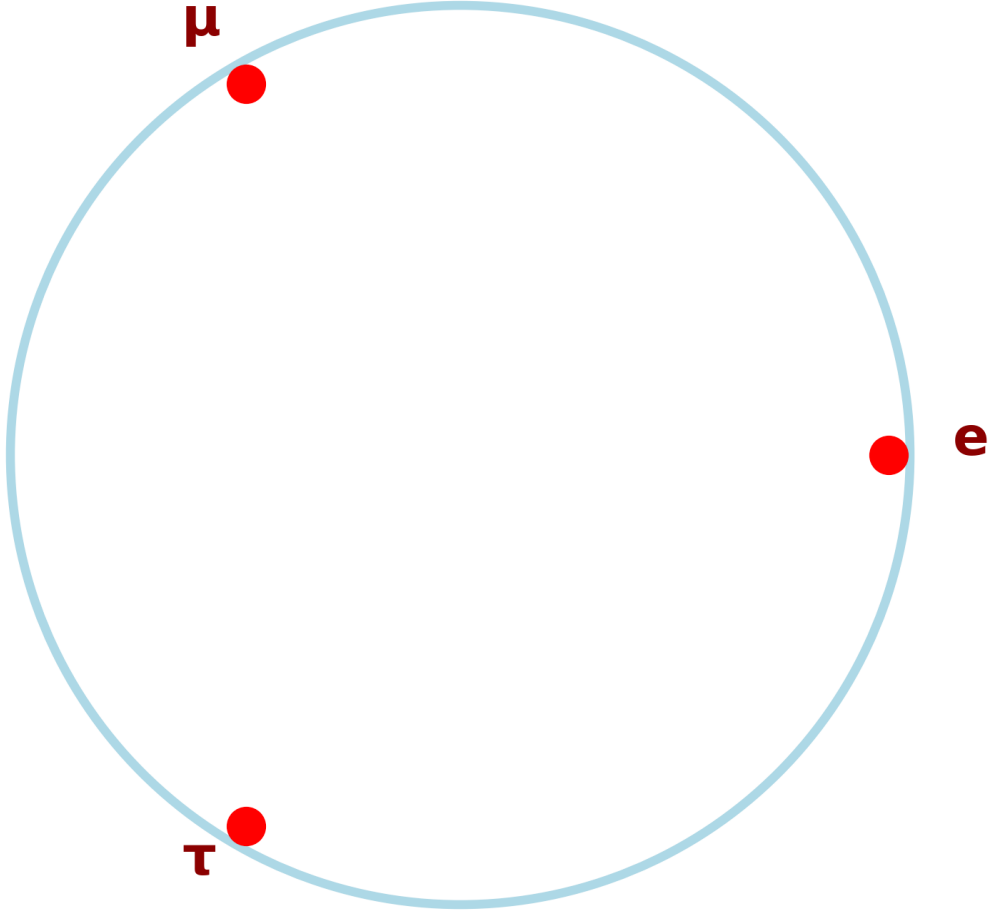


Figure 1: Schematic of the charge-fiber picture: for a fixed nonzero $U(1)$ charge, the relevant boundary degrees of freedom restrict to an S^1 orbit on the holographic screen. The Z_3 phases are equally spaced points on this circle.

Assumption 2 (Lowest-mode boundary encoding). *In the lowest-energy sector compatible with the $U(1)$ charge orbit, the boundary encoding of the square-root mass of a charged fermion is dominated by the first harmonic,*

$$\sqrt{m(\varphi)} = x + y \cos \varphi, \quad x > 0. \quad (3)$$

This is not unique, but it is the simplest nontrivial Fourier structure on S^1 and is the same phase structure used in earlier geometric discussions of Koide-like relations.

Assumption 3 (Virial partition mapped to encoding amplitudes). *The constant piece x and modulation piece y correspond to two additive contributions to the encoding “cost” (or information budget) that are identified with the virial kinetic and potential halves for an effective $1/r$ interaction:*

$$2x^2 = y^2 \quad \Longleftrightarrow \quad \frac{y}{x} = \sqrt{2}. \quad (4)$$

Assumption 3 is the only place where IAM-specific physics enters the Koide argument; it is the same $1/2$ partition emphasized in the IAM coupling $\beta_m = \Omega_m/2$ literature (Ref. [3]) and in the cosmology-focused virial-efficiency compilation (see Ref. [4]).

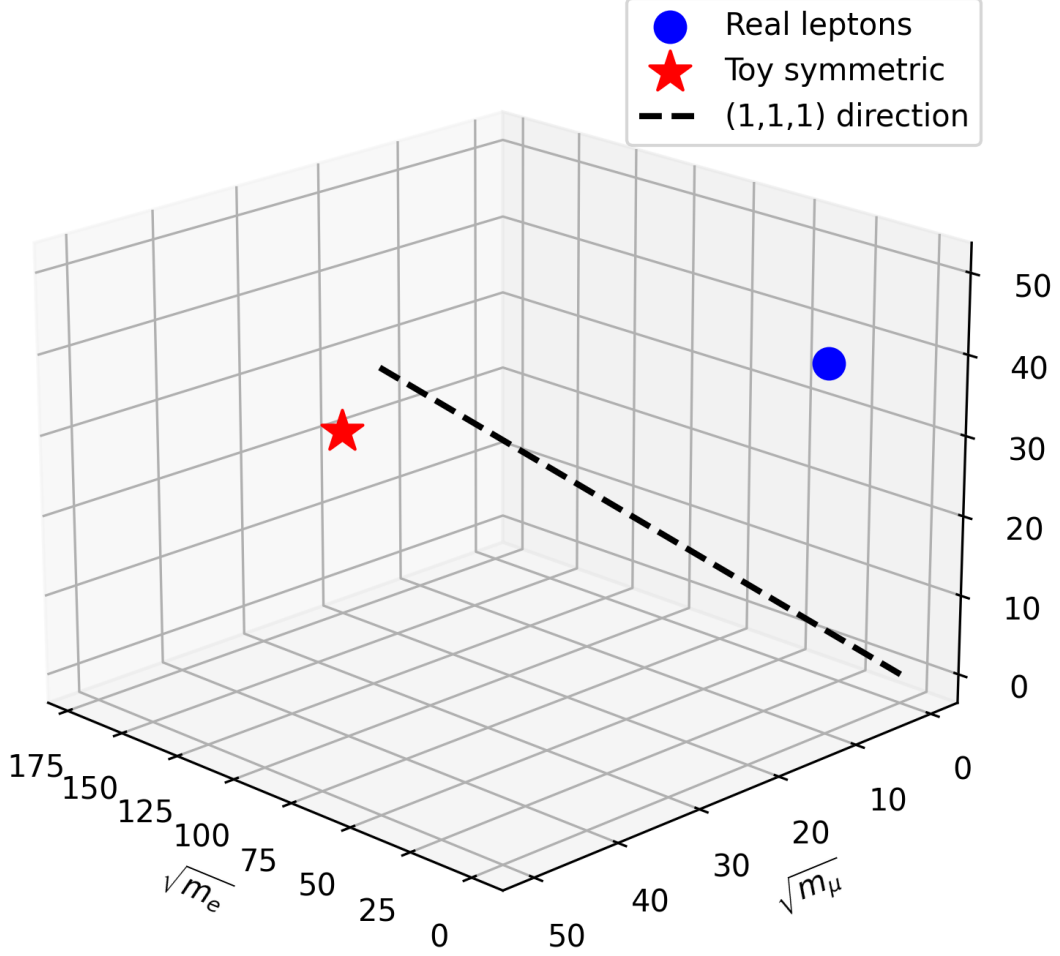


Figure 2: Geometric “flavor-space” visualization of the $(\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ triple and its Z_3 phase structure (illustrative).

The physical motivation for this identification is the following. In IAM, the virial $1/2$ partition is not a free choice: it is a mathematical consequence of the $1/r$ potential, verified analytically for both Coulomb and gravitational interactions and confirmed computationally from atomic scales (10^{-11} m) to the cosmic horizon (10^{26} m) in Ref. [3]. IAM interprets the two halves of this partition as physically distinct information channels: the kinetic half counts irreversible decoherence events (temporal encoding), while the potential half counts the configuration bits stored in spatial structure. Assumption 3 applies this same partition to the boundary encoding amplitudes: x^2 plays the role of the configuration-like (constant offset) contribution and $y^2/2$ plays the role of the event-like (modulation) contribution, with their ratio fixed by the same $1/r$ universality. This remains an assumption — the mapping from gravitational virial physics to holographic encoding amplitudes is not derived here from first principles — but it is grounded in a partition that holds without exception across every physical scale at which it has been tested.

5 Phase states and mass positivity

Definition 1 (Discrete generation phases). *For n generations, define the Z_n -symmetric phases*

$$\varphi_k = \frac{2\pi k}{n}, \quad k = 0, 1, \dots, n-1, \quad (5)$$

and masses

$$m_k = (x + y \cos \varphi_k)^2. \quad (6)$$

Physical masses require $\sqrt{m_k} = x + y \cos \varphi_k > 0$ for all k .

Theorem 1 (Uniqueness of $n = 3$ under positivity and $y/x = \sqrt{2}$). *Assume (6) with $x > 0$ and the virial constraint $y/x = \sqrt{2}$. Then the only nontrivial integer $n \geq 2$ such that $x + y \cos \varphi_k > 0$ for all k is $n = 3$.*

Proof. For $n = 2$, $\{\varphi_k\} = \{0, \pi\}$, hence $\sqrt{m_1} = x - y = x(1 - \sqrt{2}) < 0$; excluded.

For $n \geq 4$ we show $\min_k \cos \varphi_k \leq -1/\sqrt{2}$ by considering two sub-cases. *n even:* the phase $\varphi_{n/2} = \pi$ is always present, so $\min_k \cos \varphi_k = -1 < -1/\sqrt{2}$. *n odd, $n \geq 5$:* the phase nearest π is $\varphi_{(n-1)/2} = \pi(n-1)/n$. Since $(n-1)/n \geq 4/5$ for all $n \geq 5$, we have $\varphi_{(n-1)/2} \geq 4\pi/5$, and $\cos(4\pi/5) = -\frac{1}{4}(1 + \sqrt{5}) \approx -0.809 < -1/\sqrt{2} \approx -0.707$. The cosine is monotonically decreasing on $[0, \pi]$, so $\cos \varphi_{(n-1)/2} \leq \cos(4\pi/5) < -1/\sqrt{2}$ for all odd $n \geq 5$. Therefore

$$\min_k \sqrt{m_k} = x + y \min_k \cos \varphi_k \leq x + y \left(-\frac{1}{\sqrt{2}} \right) = x - \frac{y}{\sqrt{2}} = x - x = 0,$$

so strict positivity fails. Thus $n \geq 4$ is excluded.

For $n = 3$, $\cos \varphi_k \in \{1, -1/2, -1/2\}$ and $\sqrt{m_{1,2}} = x - y/2 = x(1 - \sqrt{2}/2) > 0$. Hence $n = 3$ is admissible and unique among $n \geq 2$. $\square$

6 Koide ratio as an identity for Z_3 phases

Theorem 2 (Koide identity from Z_3 and $y^2/x^2 = 2$). *Assume $n = 3$ phases $\varphi_k = 2\pi k/3$ and masses $m_k = (x + y \cos \varphi_k)^2$ with $x > 0$. If $y^2/x^2 = 2$, then the Koide ratio*

$$Q \equiv \frac{m_0 + m_1 + m_2}{(\sqrt{m_0} + \sqrt{m_1} + \sqrt{m_2})^2}$$

equals $2/3$ exactly.

Proof. For $\varphi_k = \{0, 2\pi/3, 4\pi/3\}$ one has $\sum_k \cos \varphi_k = 0$ and $\sum_k \cos^2 \varphi_k = 3/2$. Thus

$$\sum_k \sqrt{m_k} = \sum_k (x + y \cos \varphi_k) = 3x,$$

and

$$\sum_k m_k = \sum_k (x + y \cos \varphi_k)^2 = 3x^2 + y^2 \sum_k \cos^2 \varphi_k = 3x^2 + \frac{3}{2}y^2.$$

Therefore

$$Q = \frac{3x^2 + \frac{3}{2}y^2}{(3x)^2} = \frac{1}{3} \left(1 + \frac{y^2}{2x^2} \right) \stackrel{y^2/x^2=2}{=} \frac{1}{3}(1 + 1) = \frac{2}{3}.$$

$\square$

7 Numerical comparison with charged lepton pole masses

Using PDG 2022 values (as quoted in the draft), one finds a charged-lepton Koide ratio within $\sim 10^{-5}$ of $2/3$ (Table 1).

Table 1: Charged lepton masses and Koide ratio (PDG 2022 values [8]).

Quantity	Value	Source
m_e	0.51099895 MeV	PDG 2022
m_μ	105.6583755 MeV	PDG 2022
m_τ	1776.86 ± 0.12 MeV	PDG 2022
Q_{obs}	0.66666051	computed
$2/3$	0.666666...	exact

The conditional derivation above does *not* predict the absolute masses. It predicts a rigid *relation* among them once they are organized into the Z_3 structure in (1)–(5).

8 What would falsify the mechanism?

The logic has clear failure points:

- If a UV-complete holographic model does *not* reduce a charged-operator sector to an effective S^1 orbit carrying a single phase, then Assumption 1 fails.
- If the lowest-energy encoding on that orbit is *not* dominated by the first harmonic (e.g. it requires multiple comparable Fourier modes), then Assumption 2 fails; Koide need not follow.
- If the virial half-partition cannot be mapped to the two encoding contributions in a controlled statistical-mechanical argument (Assumption 3), then the numerical value $y/x = \sqrt{2}$ is not justified.

A key empirical discriminator is whether the same Z_3 phase structure appears naturally in independent microscopic constructions (e.g. models of flavor from discrete symmetries, modular symmetries, or holographic toy models), without being imposed.

9 Relation to earlier geometric observations

Foot and others emphasized geometric and phase-vector interpretations of Koide-like relations [6]. Brannen presented a phase form with $\{0, 2\pi/3, 4\pi/3\}$ phases [7]. The present note differs only in *logical packaging*: the Z_3 phase structure is here tied to a positivity constraint plus a fixed amplitude ratio $y/x = \sqrt{2}$, rather than adopted as an ansatz to fit masses.

10 Conclusions

The paper makes a narrow, referee-facing claim: *granting* three explicit assumptions (a charge-orbit S^1 , lowest-mode encoding, and a virial amplitude partition), the number of admissible positive-mass phase states is uniquely $n = 3$, and the Koide ratio follows as an exact identity.

The result is therefore not “a fit” but a conditional theorem: the difficulty is pushed to the physics of the three assumptions. That is precisely where future work must concentrate.

A Minimal comments on de Sitter holography

The standard bulk-gauge / boundary-global correspondence is on the firmest footing in AdS/CFT. Applying it to de Sitter cosmology is an active research area. IAM uses holographic-language inputs throughout its framework; this appendix does not attempt to settle those questions. Instead it clarifies the minimal role played here: only the existence of a one-parameter $U(1)$ orbit on the screen is used (Assumption 1).

B On “entropy functionals” and what can and cannot be made rigorous here

A natural question is whether an explicit entropy functional whose extremization yields both $y/x = \sqrt{2}$ and $n = 3$ can be constructed. A fully first-principles construction would require a microscopic model for boundary microstates and their energy/entropy accounting. That is beyond the scope of a short Koide note.

What *can* be made mathematically sharp is the following separation:

1. **Algebraic layer:** Given (1) and Z_n phases, the positivity theorem (Theorem 1) and Koide identity (Theorem 2) are rigorous.
2. **Model layer:** The choice of (1) and the mapping $y^2/x^2 = 2$ must come from an external physical argument (holography + virial mapping).

A referee-proof presentation therefore separates these layers cleanly (as we did above), rather than presenting the model layer as a completed derivation.

C A possible route to deriving $y/x = \sqrt{2}$ from first principles

One candidate approach would use a Weyl-squared gravitational entropy functional, which vanishes on exact FRW (since FRW is conformally flat, $C_{abcd} = 0$) and is sourced only by perturbations, naturally aligning with the IAM perturbation-sector split. A controlled derivation would require specifying a coarse-graining procedure on a spacelike hypersurface and evaluating the functional on a statistical ensemble of virialized halos; this is reserved for future work.

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